

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: # Read data
df = pd.read_csv("Advertising.csv")
df.iloc[:5, [1, -1]]
```

```
Out[2]:
```

	TV	Sales
0	230.1	22.1
1	44.5	10.4
2	17.2	9.3
3	151.5	18.5
4	180.8	12.9

```
In [3]: # Calculate minimum using minimize() function
from scipy.optimize import minimize
```

```
def obj(w):
    m, b = w
    x = df.iloc[:5, 1].values
    y = df.iloc[:5, -1].values
    pred = m * x + b

    return ((pred - y) ** 2).mean()
```

```
res = minimize(obj, x0 = [0, 0])
res.x
```

```
Out[3]: array([0.0517618 , 8.17909235])
```

```
In [4]: # Calculate minimum with normal equation
y = pd.DataFrame(df.iloc[:5, -1]) # Sales data (output)
y
```

```
Out[4]:
```

	Sales
0	22.1
1	10.4
2	9.3
3	18.5
4	12.9

```
In [5]: X = pd.DataFrame(df.iloc[:5, 1]) # TV data (input)
one = [1, 1, 1, 1, 1] # required for matrix
```

```
X["one"] = one
X
```

Out[5]:

	TV	one
0	230.1	1
1	44.5	1
2	17.2	1
3	151.5	1
4	180.8	1

In [6]: `X_transpose = np.transpose(X) # TV data transposed`
`X_transpose`

Out[6]:

	0	1	2	3	4
TV	230.1	44.5	17.2	151.5	180.8
one	1.0	1.0	1.0	1.0	1.0

In [7]: `X_transpose_X = np.dot(X_transpose, X) # Dot product of TV data transposed x TV data`
`X_transpose_X`

Out[7]: `array([[1.1086299e+05, 6.2410000e+02],`
`[6.2410000e+02, 5.0000000e+00]])`

In [8]: `X_transpose_y = np.dot(X_transpose, y) # Dot product of TV data transposed x Sales`
`X_transpose_y`

Out[8]: `array([[10843.04],`
`[ 73.2 ]])`

In [9]: `X_transpose_X_inv = np.linalg.inv(X_transpose_X) # inverse of (TV transposed data x`

In [10]: `beta = np.dot(X_transpose_X_inv, X_transpose_y) # Normal equation`
`beta`

Out[10]: `array([[0.05176182],`
`[8.17908951]])`

In [11]: `# Output from normal equation matches output from minimize() function`
`print("Minimize function: ", res.x)`
`print("Normal equation: ", beta)`

Minimize function: `[0.0517618 8.17909235]`
Normal equation: `[[0.05176182]`
`[8.17908951]]`

In []: